# Bookerfly - A Community-Based, Collaborative Reading System

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BookerFly is a reading platform which focuses on making the reading experience fun and engaging. Readers can easily interact and engage with one another through group readings, shared annotations or reading activity statistics without having to remove themselves from the reading space and breaking the immersion. In this paper, we present the design process and usability evaluation of BookerFly, and we discuss how to further expand our design in order to build a more collaborative reading experience.

CCS Concepts: • **Human-centered computing → Interface design prototyping**.

Additional Key Words and Phrases: HCI, Design, Prototype, Reading, Social

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## 1 Introduction

Since the COVID-19 pandemic, reading, especially among students, has become a more preferred leisure activity compared to others. In fact, people's reading enjoyment prior to the lockdown was positively correlated with how they enjoyed reading during and after, for both print and digital formats. (Pagani and Evgenia [12], Sun et al.[16]). In parallel, online and digital reading platforms are steadily gaining popularity among the younger generation. For instance, we have default reading apps like Apple Books[1] or Play Books[2], online publishing platforms like Wattpad[3] or Webtoon[4], and specialized tracker and review apps like Goodreads[5].

We conducted 8 story interviews, targeting young adults and students who like to read digitally, in order to identify existing issues with current reading systems. We decided to focus our system on a few recurring feedback. Multiple interviewees mentioned the "social" aspect of reading: they like engaging with others, and also rely on reviews and ratings from other users in order to decide whether or not a book is worth reading. However, in existing systems, except for this and comment sections, users usually have to leave the reading space if they want to further engage with the Community. As this breaks the immersion, we decided to target this problem as one of our priorities. We also received some interesting feedback about readers' immersion within the book and with a system: some people mentioned the possibility to annotate a book, as one would do with a physical paper

[1]https://www.apple.com/apple-books/
[2]https://play.google.com/books
[3]https://www.wattpad.com
[4]https://www.webtoons.com/en/
[5]https://www.goodreads.com/

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book. To address this, we want to provide ways for readers to interact and engage with the story directly within the context.

## 2 Literature Review

### 2.1 Community aspect of reading

Recent research converges on the idea that reading is not an isolated cognitive activity, but a socially situated practice, deeply intertwined with interaction, participation, and community. Rather than being limited to individual interpretation, reading is part of a broader social process that influences both engagement with texts and the construction of meaning (Zhang et al.[21]; Birke [2]).

A first point of consensus is the persuasive role of social interaction throughout the reading process. Social influence impacts early stages of the reading activity such as motivation and selection of readable text. It is therefore not just limited to discussions that follow reading. Readers are influenced by recommendations, shared experiences and community dynamics throughout their entire reading journey (Zhang et al.[21]; Li and Wu[9]). With the rise of digital environments, this ongoing social dimension has intensified, as platforms allow for continuous interaction before, during and after reading.

A second key point concerns the importance of interactivity and sense of social presence in the user experience. Studies by Li et al.[8] on digital social reading highlight that interactions with other readers and with the text itself contribute to a sense of belonging, which, in turn, significantly improves satisfaction and engagement. This suggests that the value of social reading platforms lies in their capacity to create an environment where readers feel socially connected even in asynchronous contexts.

The literature consistently demonstrates that social interaction is home to a deeper construction of meaning. Collaborative reading practices — such as commenting, editing, and discussion — allow readers to compare different viewpoints and refine their interpretations (Yang et al.[20]; Swann and Allington [17]). Meaning thus emerges as a collective construct arising from interaction rather than as an individual understanding of the text.

Finally, contemporary studies emphasize the role of digital communities in transforming reading into a participatory cultural practice. Online ecosystems, such as BookTok and BookTube (book-centered social media), demonstrate how readers actively contribute to the circulation, evaluation, and reinterpretation of texts (Birke [2]; Reddan et al.[14]). These environments facilitate exchanges and enable the expression of identity and the building of communities, thus reinforcing the idea that reading is inseparable from a sense of social belonging and self-affirmation.

Overall, the literature highlights three complementary dimensions of social reading: (i) its influence on the entire reading process, (ii) the role of interactivity and social presence in the user experience, and (iii) the importance of collaboratively constructing meaning within communities. These findings encourage the development of community-based reading platforms designed to foster interaction, strengthen engagement, and enable collective interpretation.

### 2.2 Usability Principles for E-book systems and their Evaluation

Electronic readers, also called e-readers are mobile electronic devices designed primarily for the purpose of storing and reading digital e-books. Thus, they usually feature e-ink instead of LCD screens, optimize battery life and portability. E-readers offer a number of advantages over traditional books, such as online access to seemingly endless numbers of texts, the ability to download many texts on a device for easy access, cost benefits, and portability (Jamali et al.[5]). In recent years, e-reading has emerged in online platforms and mobile applications, making the possession of a dedicated e-reader not necessary anymore. With how accessible e-books and digital reading has become, people only need to have access to a screen device in order to start their reading journey.

Sun et al.[16] showed that the current generation of students is *print native* rather than *digital native* , which means that, without proper support and good usable systems, it will be harder for some to profit from digital reading.

Currently, issues with electronic reading and digital reading include eye-strain, poor navigation, and restrictive usage compared to paper books (Benedetto et al. [1]; Kang et al.[7]; Sun et al.[16]). Since the birth of the first e-readers, researchers have been conducting various studies in order to create a useful and usable system that addresses as many issues as possible.

In terms of layout, studies have consistently shown that good HCI principles can facilitate active reading the same way paper does by implementing HCI design principles, e.g. consistency with the real world, completeness of the system and system conventions (Pearson et al.[13]; Wang and Huang [19]). In addition, e-reading systems should implement magnification with adaptable page numbers and legible text. In this regard, Sheedy et al.[15]'s study tested reader's enjoyment and reading comprehension with 4 different fonts, and found that Verdana and Arial were the most legible, with a minimum font size of 10pt. This study also revealed that while nicer page layout aesthetics did not improve reading performance, they were preferred by readers and could improve their mood.

Multiple studies across the literature mention the need for annotations. In fact, annotations are a popular form of interaction with written text, and a core feature of paper books. Ideally, they should be flexible and interactive to improve active reading[18]. Pearson et al.'s study also mentioned the importance of usable bookmarks: they should be easily accessible, and facilitate navigation across the interface.[13]

### 2.3 Digital annotations

Annotating is a useful technique, allowing people to add information to existing documents. It is alrady a long established process when working with paper-based materials. As an artifact, annotations can provide alternative interpretations of a piece of content that may lead a reader to recontextualize the events of a story (Glover et al.[4]).

Glover et al.[4] ran a study on university students and academics in order to analyse their style of annotations. The most common use was *highlighting*, in order to quickly and neatly identify key information. These highlights are a generic form of annotation with commonly admitted connotations, whereas other types of annotation, such as notes and symbols, may only have explicit meaning for the original annotator. This supports the need of different 'access levels' for each individual or groups of annotations, with smooth transitions between each (Glover et al.[4]; Marshal [10]).

The need for public and private annotations has been consistently supported throughout the literature. Some systems have expanded on them, turning those annotations into fully interactable artifacts: for instance, other users being able to respond to public annotations, or even save them for later. (Jiun-Chi Jan et al.[11]; Chih-Ming Chena et al.[3]).

Marshall's study[10] on the form, function and community status of annotations suggests a set of design implications for digital readers:

- (i) Annotations should be visually distinguishable from the source document. On paper, they are highly individual in form: digital annotations should respect this fluidity.
- (ii) Reading systems should enable users to develop personalized annotation systems, in which they could use colors and symbols to create meanings.
- (iii) Despite being fully integrated within the reading process, annotations should be designed so they interrupt reading as little as possible.

However, excessive and poor-quality annotations within the text might cause information overload and divert the reader's attention away from the main content (Jiun-Chi Jan et al.[6]). The increased cognitive load can then ultimately reduce the effectiveness of collaborative annotations in promoting reading comprehension.

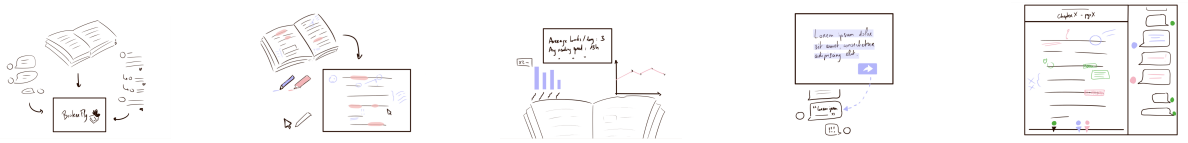
## 3 Design Methodology

In order to clearly define the core features of our system, we used the Delphi Procedure to clarify each of them. First, each member of our group made a personal list of 5 to 10 features they deemed important for our platform. Then, we sent them to a panel of experts (here, the members of Group 3) in order to synthesize them. They compiled our everything and transformed them into a list of 11 statements. Then, we were sent the list back for us to review it, and eventually refine it further. After that, they compiled our changes again by features, and sent each of us a form to rank everything in order of importance (1 being the most important and 11 the least). A clear consensus emerged quickly: our top 3 features were group reads, annotations and snippets with respective scores.

## 4 Prototype

To put our concept into practice, we began by designing a prototype using paper prototyping techniques. Based on the results of the Delphi Procedure (see previous section), we established a clear list of Bookerfly's core features, which guided our design decisions.

We first sketched initial ideas for the overall layout and visual structure of the interface. To ensure usability and familiarity, we drew inspiration from existing platforms when designing the main page. This approach allowed us to rely on recognizable navigation patterns while integrating Bookerfly's collaborative features.



(a) Messaging (b) Annotations (c) Statistics (d) Snippets (e) Group reads

Fig. 1. Early sketches of Bookerfly

Following the paper prototyping phase, we developed a functional high-fidelity prototype using HTML, CSS, and JavaScript. The goal of this implementation was not to build a production-ready system, but to simulate realistic interactions and evaluate usability, navigation, and feature discoverability.

The application is structured as a multi-page web prototype, with separate views for the main sections: Library, Reader, Chats, Snippets, and Profile (in addition to a mini Study Session interface that was only used for testing purposes). These pages are connected through a persistent top navigation bar, ensuring consistency across the experience. A central data file (data.js) acts as a mock backend, storing all application content such as books, text passages, chat messages, user profile data, and statistics. This decision was made to avoid the need for a server or database while still enabling a realistic simulation of the system.

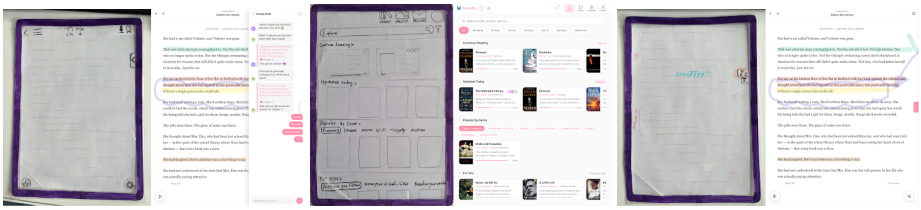


Fig. 2. From paper prototyping to implementation

User interactions and UI behaviour are handled through shared utilities defined in app.js. These include common components such as notifications, the navigation bar, dark/light mode management, and reusable rendering functions (e.g., book cards). State persistence is achieved using the browser's localStorage, allowing user actions-such as messages, annotations, or profile edits-to persist across sessions.

Each feature is implemented with a strong focus on interactivity:

- Reading can take place in multiple modes (individual or group-based). Users can bookmark chapters and interact with the text through selection-based actions such as highlighting, adding notes, or saving snippets.
- To add annotations, users can activate a dedicated annotation mode by double-tapping the screen. This interaction temporarily removes interface elements and reveals a pen menu, including a pen, highlighter, and eraser, allowing users to draw or write directly on the page.
- For the group reading feature, a group chat is embedded directly within the reading interface. This design enables readers participating in the same session to discuss passages or share snippets without leaving the reading environment. The chat is accessible via an arrow on the side of the screen, allowing users to expand or collapse the panel as needed, thereby maintaining readability while keeping discussions readily available.
- The Snippets system allows users to extract and manage quotes, reinforcing the social and reflective aspects of reading.
- The Profile and statistics views simulate different types of personalised data tracking and visualisation (e.g. number of books read)

Overall, this implementation prioritises functional realism and testability over completeness. It enabled us to evaluate how users interact with Bookerfly's core features in a controlled environment.

## 5 Usability Evaluation

### 5.1 Participants

We conducted a usability study on 8 participants (4 women and 4 men, recruited via word of mouth on campus) in order to test the discoverability and usability of BookerFly's core features – group reads, annotations and snippets – as well as the ease-of-navigation of the overall interface. Participants were all in their early twenties. They were all students used to reading digitally, with a group average of 5 hours per week. In order to avoid possible bias, we excluded participants from our prior comparative study.

### 5.2 Material

All participants used the same IPad tablet and laptop (MacOS) to navigate the interface, to avoid hardware bias. Those devices were provided by us. The study was run on campus grounds, in an isolated room so there would be no external disturbances.

### 5.3 Protocol

For this experiment, we decided to test how intuitive and easy BookerFly was for first-time users. The main hypothesis around which we articulated this study was "Participants are able to discover and use core features without external help and within a reasonable time period"

We started by giving them an introduction to our system, explaining the core concepts mentioned above. They were then provided with a list of tasks to accomplish in order. Finally, participants were left to freely explore the system, with a limit of 10 minutes. The entire experiment lasted around 20 minutes. Below is the list of steps participants had to follow:

(1) Open an ongoing Group Read from the home page.
(2) Find the embedded chat in the Group Read.
(3) Flip a few pages .
(4) Open the Annotation Mode and annotate the page. Such annotations have to include: Drawing Annotations and Text Highlights.
(5) Show and hide other users' annotations.
(6) Select an excerpt of text, and share it as a Snippet to the relevant group chat.
(7) Free exploration of the system.

### 5.4 Data collection

We ran this study using the think aloud method: with their consent, participants' voices were recorded and transformed into text for us to analyse. In addition to that, we also observed their behaviours and expressions and took notes of any significant insights. Finally, we collected logs of participants' clicks and their respective timestamps. After the study concluded, we collected their demographic information and let them fill a NASA-TLX questionnaire to approximate mental load. Afterwards we used a Google Forms questionnaire to gather more open-ended feedback and suggestions. Such questions included: "Overall, did you find the features discoverable?" or "What was your favorite feature? Why?" We collected and anonymized each participant's data using unique user IDs (P1 to 8).

## 6 Results

### 6.1 NASA-TLX results

With an average of 3.5 out of 7, our system was deemed neutral in terms of metal demand, neither simple nor complex to use. The time demand, in spite of that, was slightly higher than we would have expected, averaging at 3.88. However this was probably caused by the context of the experiment, with participants being observed at all times. Overall, users were satisfied with their performance, with a low average frustration and workload score.

### 6.2 Observations and open-ended questionnaire

Participants said the interface was easy to navigate. 5 out of 8 said the features were easily discoverable, while the remaining 3 mentioned that snippets and annotations were difficult to find. 3 participants chose snippets as their favorite feature, stating that they liked being able to seamlessly share excerpts to group chat in a neat format. 2 participants chose annotations and, specifically freehand ones, because "the possibility for others to remove my drawings with a toggle makes it so I don't feel bad covering up words" (P3). 2 more picked activity statistics as their favorite feature, and the last participant chose the snippets spoiler warning.

In terms of critics, annotations were deemed "hard to find", which we noticed as several participants struggled to understand how to activate annotation mode. One participant also reported that the help button was difficult to find, as it could only show up on the homepage and nowhere else, and suggested having it on all screens instead. We were also surprised to notice that 5 out of 8 participants mistakenly thought that group reads were

stored in the group chats. Despite the fact that we described group reads as normal books, as well as naming a reading tab "Groups", participants consistently associated the feature with chats instead. Overall, all participants said they liked Bookerfly. P1 declared: *"I would actually love to have that in real life"*, and P5 mentioned they *"feel it [could] be really interesting, and [could] also make the whole reading process much more enjoyable"*.

## 7 Discussion

Based on the results of our study, we can confidently say that our proposal is relevant for enhancing reading experience through a community lense. The positive feedback on our core features – group reads, snippets and annotations – confirms that there's a demand for more social reading systems: participants appreciated group reads, found the social features engaging, and felt that annotations and snippets brought forth new ways of interacting with stories.

This study also highlighted persistent issues with our design: lack of discoverability for certain features, confusion between some icons and some interactions not being clear at first glance. These observations highlight the importance of usability in addition to functionality. We also noticed that users required some time to understand the system, suggesting the need for better onboarding and more intuitive interactions.

Following our own observations and participants' suggestions, we decided make some changes to our system:

- Render the *Help* button accessible from everywhere. That way, new users can easily find what they're looking for without breaking their stride.
- *Add more discoverability to features.* Our prototype lacked enough visibility onto this aspect.
- *Supporting working typed annotations on both devices.* Since the Apple menu kept overriding BookerFly's own menu while we were testing our system, we had to switch to laptop in order to access selection highlight, snippets tool and typed annotation panels.
- *Better onboarding for new users and clearer interactions*, in order to facilitate navigation and improve the usability experience.

It is important to note that the results of this study might be a bit biased, as participants were directly recruited at our university. This could have potentially influenced their feedback and limited the diversity of perspectives.

## 8 Conclusion

We introduced BookerFly, a reading platform focused on collaborative reading and interactions between users. We saw through our user study that our core social features – *Group reads*, *Shareable snippets* and *Public and private annotations* – were greatly appreciated and noticeably enhanced participants' reading experience.

BookerFly can help users engage with each other and with stories through original ways without breaking their immersion within the system. With a few more improvements, we believe our system can contribute to the design of more social and immersive reading platforms.

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